



DS3

Data Science for
Developing Scholars in
Down Syndrome Research

Exercises in Data Cleaning and Exploratory Data Analysis

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Data Science for Developing Scholars in
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Links for this session

Course materials

[DS3 2026 repository on GitHub](#)

[DS3 Tidy Data Exercise](#)

[DS3 Intro to exploratory data analysis](#)

Tidyverse and data visualization

[Tidyverse](#)

[ggplot2](#)

[ggplot2 book](#)

<https://www.data-to-viz.com/caveats.html>

<https://r-graph-gallery.com/index.html>

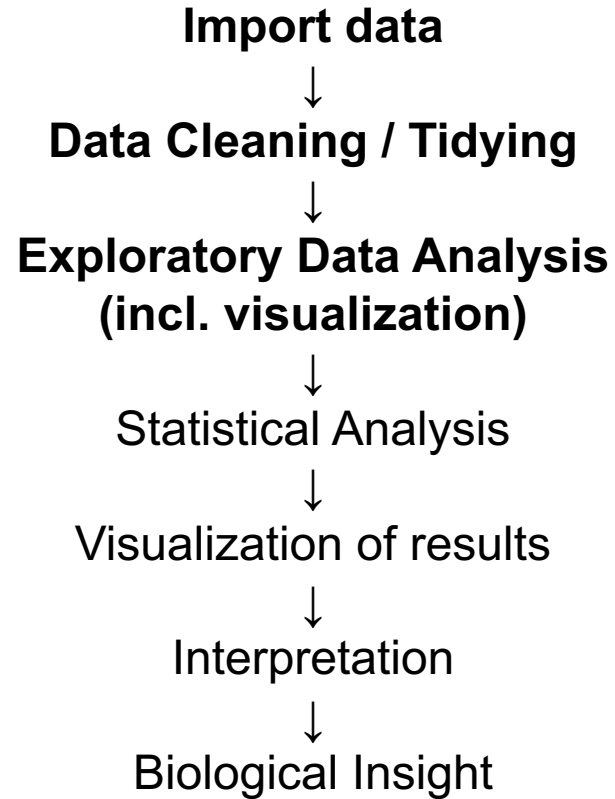
Today's Goals

By the end of today you should be able to:

- Open and work within an R Project
- Read data into R
- Recognize tidy vs. messy data
- Use basic dplyr verbs
- Create simple ggplot figures
- Follow a reproducible analysis workflow

Most of today's work will be performed directly in R scripts rather than slides.

The Data Science Workflow



Today's exercises focus on data import, cleaning, exploration, and preliminary visualization

Hands on:

- 1. Tidy data exercise**
- 2. Intro to exploratory data analysis**

How we will use Git and GitHub

GitHub repository: https://github.com/DS3-Course/DS3_2026

**Simple workflow for accessing and updating
course materials for DS3 afternoon sessions**

- **Clone** the course repository once at the beginning:
`git clone https://github.com/DS3-Course/DS3_2026.git`
- **Pull** updates each day to access new content:
`git pull`
Alternative: direct download from github
- **Open** the appropriate `.Rproj` file for each lesson in RStudio.
- Follow instructions in README.md files and R scripts.